

SQL Sales & Customer Order Analysis Project

This document explains the SQL-based sales and customer order analysis project performed using retail datasets. The project focuses on generating business insights using SQL queries and relational data analysis techniques.

1. Project Objective

The objective of this project was to analyze retail sales and customer order data using SQL to identify business trends, customer behavior, profitability, and product performance.

2. Database Structure

Table Name	Description
Customer_Order_Data	Contains customer order information and sales transaction details.
Sales_Detail_Data	Contains product-level details such as quantity, category, and profit information.

3. SQL Concepts Used

- SELECT statements
- Filtering using WHERE clause
- GROUP BY aggregation
- ORDER BY sorting
- Aggregate functions such as SUM(), AVG(), COUNT()
- Joins between multiple tables
- Business KPI calculations

4. Business Questions Solved

- What is the total sales revenue?
- Which products generated the highest sales?
- Which customers contributed the highest revenue?
- Which categories generated the highest profit?
- What are the monthly sales trends?
- Which products generated losses?

5. Key Insights Generated

- Top-performing product categories contributed significantly to revenue.
- Certain products generated negative profit due to discounts.
- Top customers contributed major sales revenue.

- Sales trends varied across different periods.
- Profitability differed across products and categories.

6. Project Files Included

File Name	Purpose
Customer_Order_Data.csv	Customer order dataset
Sales_Detail_Data.csv	Sales detail dataset
Sales_Analysis_Queries.sql	SQL business analysis queries

7. Conclusion

This SQL project demonstrates practical SQL skills, business-focused analysis, and the ability to generate meaningful insights from retail sales datasets.